How it will be helpful instead of using regular regression test with playwright and pytest?

Regular Playwright + pytest tests are best for **known, exact behavior**. Adding `qwen2.5-0.5b` and `qwen3-vl-2b-instruct` helps when the expected result is **semantic, visual, or difficult to express as a fixed selector.** Your uploaded tests already cover exact checks such as page title, navigation, and a todo count changing to `1`.[1]

Regular tests vs AI-assisted tests

Area	Regular Playwright/pytest	AI-assisted regression
Best for	Deterministic functionality	Flexible content and visual checks
Example assertion	Todo count contains `1` [1]	Screenshot shows the newly added todo
Speed and cost	Fast, cheap, highly repeatable	Slower and dependent on model output
Reliability	Very high	Good for broad checks, but can produce false positives/negatives
Maintenance	Breaks when selectors/text change	More tolerant to wording and layout changes
Coverage	What you explicitly script	Can catch unexpected visual or content regressions

Where AI adds value

1. Visual regression without pixel-perfect baselines

Traditional visual testing often compares screenshots pixel-by-pixel or against stored baseline images. That can fail because of minor rendering differences, fonts, animations, or intentional cosmetic changes.

A vision model can answer broader questions:

```
Does this screenshot show the expected heading?
Is there an error banner?
Is the main call-to-action visible?
Does the layout look broken or overlapping?
```

For example, instead of asserting an exact CSS selector, you can screenshot the Playwright homepage and ask `qwen3-vl-2b-instruct` whether the main heading is visible and no error state is shown.

2. Checking meaning, not exact text

A conventional test might assert:

```
expect(page.get_by_text("Installation")).to_be_visible()
```

That fails if the site changes the wording to “Setup” or “Install Playwright,” even though the page still serves the same purpose.

A text model can check intent:

```
Does this page explain how to install or get started with Playwright?
Answer YES or NO.
```

This is useful for content-heavy pages, documentation, marketing pages, and localized sites.

3. Reducing selector maintenance

UI redesigns often change class names, test IDs, and DOM structure. A visual check can still validate the user-visible outcome even when selectors change.

For instance, after a redesign, your Playwright test may fail because the todo-count test ID changed. A screenshot-based AI test can still ask:

```
Does this TodoMVC screenshot show one active todo item?
```

That makes the check more resilient to implementation changes.

4. Catching issues conventional assertions miss

Playwright is excellent at checking that an element exists, but it does not naturally judge whether the page **looks correct**. AI visual tests can help detect:

- Missing images or major blank areas
- Overlapping or clipped UI elements
- Unexpected error messages
- Wrong page rendered after navigation
- Broken layout after a CSS or dependency change
- Missing important content in a screenshot

These are supplementary checks, not replacements for functional assertions.

What AI should not replace

Keep deterministic Playwright tests for anything that must be exact:

- Login works
- Add-to-cart updates the cart count
- Form validation shows the required error
- Checkout completes successfully
- API responses contain expected fields
- Critical text, prices, and legal content are exact

Do not use an LLM to assert something like “the price is exactly $19.99” or “the order ID is 12345.” Use normal assertions for those.

Practical rule

Use this layering:

1. **Playwright + pytest**: Verify the workflow works exactly.
2. **qwen2.5-0.5b**: Verify page text has the right meaning or summary.
3. **qwen3-vl-2b-instruct**: Verify the rendered screenshot looks correct and contains expected visual elements.

For your current examples, retain the exact todo-count assertion from your document, then add a VLM check that confirms the newly created todo is visible in the screenshot. That combination gives you both reliability and broader regression coverage. [1]

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1. simple-tests.md